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Raghu Raman

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Transparency in research: An analysis of ChatGPT usage acknowledgment by authors across disciplines and geographies

Raghu Raman

Amrita School of Business, Amrita Vishwa Vidyapeetham, Amritapuri, Kerala, India

ABSTRACT

This investigation systematically reviews the recognition of generative AI tools, particularly ChatGPT, in scholarly literature. Utilizing 1,226 publications from the Dimensions database, ranging from November 2022 to July 2023, the research scrutinizes temporal trends and distribution across disciplines and regions. U.S.-based authors lead in acknowledgments, with notable contributions from China and India. Predominantly, Biomedical and Clinical Sciences, as well as Information and Computing Sciences, are engaging with these AI tools. Publications like “The Lancet Digital Health” and platforms such as “bioRxiv” are recurrent venues for such acknowledgments, highlighting AI’s growing impact on research dissemination. The analysis is confined to the Dimensions database, thus potentially overlooking other sources and grey literature. Additionally, the study abstains from examining the acknowledgments’ quality or ethical considerations. Findings are beneficial for stakeholders, providing a basis for policy and scholarly discourse on ethical AI use in academia. This study represents the inaugural comprehensive empirical assessment of AI acknowledgment patterns in academic contexts, addressing a previously unexplored aspect of scholarly communication.

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Ethics in publishing; fields of research; social science research ethics; research integrity; science policy

Introduction and related work

The advent of Artificial Intelligence (AI) tools has revolutionized numerous industries, academia being no exception. AI technologies, such as the Generative Pre-trained Transformer (GPT), a tool developed by OpenAI, have transformed how researchers write and construct academic papers (Alattar and McDowell 2023; Salvagno, Taccone, and Gerli 2023). The utility of this tool lies in its capacity to generate human-like text, providing valuable assistance in academic writing (Alkaissi and McFarlane 2023; Chen 2023; Macdonald et al. 2023). Yet, despite their increasing usage, the transparency in acknowledging these tools in research papers is a topic that has not been

sufficiently addressed, raising crucial questions about academic ethics and disclosure.

ChatGPT, a large language model (LLM) developed by OpenAI (OpenAI 2023), is a powerful tool that can more often generate fluent, and informative text that is human-like. It was trained on diverse internet text and employed machine learning techniques to produce contextually relevant, syntactically correct, semantically rich text. A distinguishing feature of ChatGPT is its capacity to comprehend nuanced prompts and generate comprehensive responses, making it an invaluable assistant for academic writing. It can help researchers navigate the intricacies of scientific writing, thereby improving the efficiency of writing processes. For instance, researchers can input an outline or keywords into ChatGPT, and it can assist in formulating comprehensive paragraphs. Similarly, it can enhance readability by suggesting rephrasing for complex sentences or jargon-laden passages. Researchers have used ChatGPT to write various scientific papers, including review articles, research articles, and grant proposals (Lund et al. 2023).

The evolution and integration of AI in various sectors have been a popular area of research in the past decade. In academia, this interest has led to the emergence of a body of literature examining the application of AI in academic writing (Thorp 2023). Other AI tools used include Grammarly, ProWritingAid, and QuillBot. However, ChatGPT is unique in its ability to generate more human-like, fluent, and informative text. Early indications are that while AI tools like ChatGPT can possibly enhance the efficiency and accessibility of scientific writing, they do not replace the intellectual engagement, critical thinking, and creativity integral to scientific exploration. Hence, AI must serve as an aid rather than a substitute in writing, thereby giving rise to the essential question of maintaining transparency and ethical considerations in acknowledging the usage of such tools.

Transparency in research has always been a cornerstone of academia (Wilhite & Fong, 2012). The open declaration of methodologies and tools used in the research process strengthens trust in scientific findings and allows for reproducibility, an essential aspect of scientific progress (Munafò et al., 2017). However, the proliferation of AI tools such as ChatGPT in academic writing necessitates discussing transparency and ethics. The principles of openness, reliability, and honesty are cornerstones of academic research, and generative AI tools must not compromise these. Researchers who utilize generative AI tools in their writing must disclose this usage to uphold the integrity of their work. This call for transparency aligns with the increasing emphasis on open Science and the global trend toward data and methodology accessibility.

The lack of acknowledgment of AI tools could lead to a breach of these core academic principles, thereby emphasizing the importance of studying such patterns. Increasingly, scientific journals insist on including data

availability statements to enhance transparency and reproducibility in research. Author guidelines delineate which parts of the research procedure should be disclosed for community assessment, feedback, application, and further development. Scientists acknowledge the importance of being transparent, open, and replicable (Nosek et al. 2015).

Mentioning Language Learning Models (LLMs) in the acknowledgments has received academic support. Jenkins and Lin (2023) advocate acknowledging the use of LLMs in this section. Nature's Editor-in-Chief echoes this sentiment, emphasizing the need to document LLM usage in either methods or acknowledgments (Stokel-Walker 2023). According to Katz et al., 2020, standard academic practice requires that research tools be described in the body of the text, along with appropriate citations. Blanco-Gonzalez et al. (2023) serve as an example, detailing their use of ChatGPT within the main text. However, this approach presents challenges regarding article findability or paywalled articles. Such limitations can hinder transparency and openness, especially if the use of LLMs is underreported.

An article by Mohammad Hosseini et al., (2023) delves into the ethical dilemmas surrounding using large language models like ChatGPT in academic writing, specifically addressing responsible authorship and disclosure concerns. It critiques existing bans on LLMs in journals such as Science, arguing that such prohibitions are impractical and may incentivize undisclosed usage while also dismissing the notion of acknowledging LLMs in the acknowledgments section on ethical and legal grounds. The article provides actionable recommendations for researchers, suggesting that they should transparently disclose the use of LLMs in the introduction or methods section, use in-text citations and references for these tools, and submit interactions with LLMs as supplementary material.

While existing literature acknowledges the use of AI tools in academic publications, it largely focuses on ethical and procedural considerations. There is a research gap in empirical studies that scrutinizes the actual practices of acknowledging AI tools, such as ChatGPT, across multiple variables like geography, authorship, journal affiliation, and institutional context. This study is the first attempt at an empirical investigation into the acknowledgment patterns related to AI tools, thereby exploring trends in the usage of generative AI tools in scientific research. Furthermore, this research sets a benchmark for further exploration into the use of generative AI tools in scientific research.

Methods

We built our dataset by following the PRISMA-P 2015 framework (Moher et al. 2015), which provides a guide for carrying out a systematic review (Figure 1) in five steps: 1) pinpointing the research questions; 2) creating a search strategy that includes the databases to be used, search terms, and what to include or exclude; 3) searching the existing literature; 4) filtering the gathered literature based on their title and abstract and then full text; and 5) analyzing the chosen materials. We used the Dimensions database to find all publications that matched our research topic, covering 2022 to 2023 (up until 16 July 2023). Following the PRISMA-P shown in Figure 1, we identified 1307 publications using Dimensions API and finally selected 1226 publications for analysis.

The keywords were carefully chosen to capture the most relevant publications in the field of Generative Artificial Intelligence, including various terminologies and acronyms that are standard in the literature, such as “Generative AI,” “ChatGPT,” “large language models,” and “OpenAI.” The Dimensions database allows targeted keyword searches across various sections of academic papers, thereby enhancing the specificity and relevance of the search. In this context, the following keywords were selected:

- “ChatGPT,” “Chat GPT,” “Chat-GPT:” These terms target publications specifically focusing on the ChatGPT model.
- “Generative AI,” “Generative Artificial,” “Generative Content:” These keywords aim to capture the broader field of generative AI technologies.
- “Large language models:” This term includes publications on sizable language models, of which ChatGPT is an example.
- “Artificial Intelligence and Generative Content:” This combination ensures that the AI technology discussed is generative in nature.
- “AI-assisted:” To include publications where AI serves a supportive function.
- “OpenAI,” “GPT-3,” “GPT-4:” These terms target publications linked to the parent company and specific versions of the model.

Thus, the selected keywords aim to provide a comprehensive yet targeted set of publications for analysis, aligned with the research questions posited in the study.

The search query used in the Dimensions database is shown below.

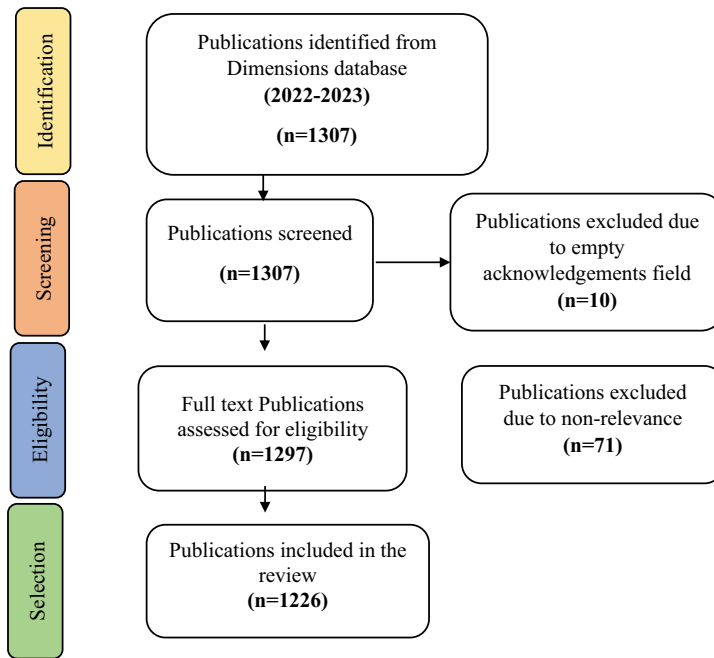


Figure 1. PRISMA-P 2015 framework.

((acknowledgments:("chatgpt" OR "chat GPT" OR "chat-GPT") OR
 acknowledgments: ("Generat* AI" OR "Generat* Artificial*" OR "Generat*
 Content") OR
 acknowledgments: ("large language model*") OR
 acknowledgments: ("Artificial Intelligence" AND "Gener* Content") OR
 acknowledgments: (("Artificial Intelligence" AND assisted) OR "AI-
 assisted") OR
 acknowledgments: ("openai") OR
 acknowledgments: ("gpt-4" OR "gpt-3*")
) OR ((openai.com) OR (openai.com/*gpt*) OR (openai.com/*/*gpt*))
 AND
 years = 2022, 2023

Our approach was focused on categorizing text from the acknowledgment section into four defined categories:

- (1) **Explicit Content Generation Acknowledgment:** This category consolidates acknowledgments explicitly stating that ChatGPT generated either a portion or the entirety of the publication's content. It accounts for instances where authors transparently admit the direct content generation role of ChatGPT.

- (2) **Assistance and Collaboration Acknowledgment:** This category encapsulates acknowledgments indicating various assistance types provided by ChatGPT, such as editing, literature review proofreading, language refinement, rewording, revising, polishing, data analysis, and more. It also includes cases where ChatGPT is cited as part of a collaborative effort involving human authors and the AI tool.
- (3) **OpenAI and ChatGPT Role Acknowledgment:** This category comprises acknowledgments attributing a specific role to OpenAI and ChatGPT in the writing process. Authors in this category express gratitude to OpenAI, credit the model, and acknowledge its role, explicitly appreciating the availability and impact of the AI tool on their work.
- (4) **OpenAI Acknowledgment in the references:** This category consists of publications that explicitly have a citation to (openai.com) OR (openai.com) OR (openai.com/*gpt*) OR (openai.com/*gpt*) in the references section.

Following establishing these categories, two independent coders were assigned to read each acknowledgment text and assign it to one or more of the three categories, as applicable. This assignment was based on the degree of alignment between the acknowledgment's text and the descriptive boundaries of each category. Please note that some acknowledgments are not straightforward and require a bit of interpretation, which is subjective.

To ensure the reliability and consistency of the categorization process, we used a third coder as an adjudicator. This coder was used whenever the first two coders disagreed on a particular categorization. The third coder's role was to review the given acknowledgment and decide which was considered final.

Each publication is analyzed based on the field of research, the corresponding author's geographical location, journals, and institutional affiliation. For temporal analysis, we sort the data according to the publication month.

Results and discussion

We present the findings divided into four main subsections:

- (1) **Field of Research differences** – In this section, we will present results that showcase how the acknowledgment of AI tools like ChatGPT varies across different fields of research. It will discuss the fields where these tools are most commonly acknowledged and those where they are relatively less mentioned.
- (2) **Geographical distribution of publications, institutions, and authors** – Here, we present an analysis of how acknowledgment practices vary based on the geographical location of authors and institutions.

- (3) Journal patterns – We look at the distribution of publications based on different journals, including preprints.
- (4) Temporal trends – Here, we show early trends regarding acknowledging AI tools like chatGPT in academic research by authors.

Fields of research differences

The Dimensions database uses the Australian and New Zealand Standard Research Classification (ANZSRC), which is structured into 22 broad Fields of Research (FoR) divisions, enabling fine-grained classification of research activities. A single publication may be cross-referenced under multiple Fields of Research (FoR) categories within the ANZSRC system. Raman et al. (2023) also employed Fields of Research (FoR) categorization in their examination of ChatGPT publications. Their analysis revealed that publications classified under “Information and Computing Sciences” and “Biomedical and Clinical Sciences” garnered the most mentions on social media platforms.

Table 1 shows the top ten FoRs based on the number of publications acknowledging ChatGPT usage.

Biomedical and Clinical Sciences leads with the most publications (410), with an average of 5.4 citations per publication, reflecting a moderate impact within the research community. Information and Computing Sciences has 278 publications and a higher mean citation rate (4.5), indicating a reasonable degree of influence. Despite having fewer publications (150), Health Sciences shows a high citation count (2090) and the highest mean citations (13.9). This suggests that publications in this field have a substantial influence. Engineering, with 111 publications, shows moderate acknowledgment of the usage of ChatGPT.

Education, Commerce Management, Tourism and Services, Psychology, and Language Communication and Culture are the other fields with relatively few publications yet indicate that ChatGPT’s usage is not confined to physical, health, and engineering sciences alone. Education, notably, despite having only 96 publications, has a significantly high mean citation rate of 13.3.

In summary, these data reveal widespread use of ChatGPT across diverse fields of research, with varying levels of impact as measured by citations.

Table 2 shows how acknowledgment of ChatGPT’s usage is categorized across five major fields of research (FoR). We have selected the top four fields of research from Table 1 and added Psychology as the fifth one to compare with a social sciences field of research.

Biomedical and Clinical Sciences, with 410 publications, leads to the acknowledgment of ChatGPT. Of these, 24% used it for explicit content generation, 41% for assistance and collaboration, and 37% acknowledged

Table 1. Number of publications acknowledging the usage of ChatGPT across various fields of research.

Name	Fields of Research	Publications	Citations
Biomedical and Clinical Sciences	32	410	2214
Information and Computing Sciences	46	278	1254
Health Sciences	42	150	2090
Engineering	40	111	264
Education	39	96	1274
Biological Sciences	31	57	50
Commerce, Management, Tourism and Services	35	44	303
Psychology	52	38	102
Language, Communication and Culture	47	36	170
Chemical Sciences	34	32	76

Table 2. Number of publications acknowledging usage of ChatGPT categorized by various fields of research.

Field of Research	Total Publications	Explicit Content Generation	Assistance and Collaboration	OpenAI and ChatGPT Role	Acknowledgement in the References
32 Biomedical and Clinical Sciences	410	99 (24%)	169 (41%)	152 (37%)	216 (53%)
46 Information and Computing Sciences	278	16 (6%)	38 (14%)	41 (15%)	229 (82%)
42 Health Sciences	150	34 (23%)	58 (39%)	53 (35%)	86 (57%)
40 Engineering	111	38 (34%)	55 (50%)	52 (47%)	51 (46%)
52 Psychology	38	6 (16%)	15 (40%)	11 (29%)	20 (53%)

the role of OpenAI and ChatGPT, with 53% citing them in the references. Information and Computing Sciences, with 278 publications, shows lower utilization for explicit content (6%) but a significant acknowledgment rate of 82% in the references.

In Health Sciences, among 150 publications, 23% used ChatGPT for explicit content generation, while 39% leveraged it for assistance and collaboration. A significant 57% of the publications acknowledged ChatGPT and OpenAI in their references. In Engineering, with 111 publications, the percentages are evenly distributed: 34% for explicit content, 50% for assistance and collaboration, and 46% for references.

Psychology, included as a social sciences field for comparison, comprises 38 total publications. The data reveals 16% for explicit content generation, 40% for assistance and collaboration, and 29% acknowledging the role of OpenAI and ChatGPT, with 53% including them in references.

The analysis indicates a consistent trend of high acknowledgment for assistance and collaboration across all fields. The highest acknowledgment rate is observed in Information and Computing Sciences at 82%, whereas Psychology also exhibits substantial utilization at 53%, indicating that ChatGPT's applications also cover social sciences.

Tables 3–7 shows the categorization of three highly cited publications acknowledging the usage of ChatGPT from five fields of research.

Geographical distribution of publications, institutions, and authors

Table 8 shows the Top ten countries based on the number of publications acknowledging the usage of ChatGPT. The United States stands out, acknowledging the highest usage of ChatGPT (407). This indicates a strong inclination toward using generative AI tools in academic writing in the U.S. Notably, China and India, both emerging economies, also show substantial usage with 92 and 75 publications, respectively. This suggests that despite the differences in economic status, AI tools like ChatGPT are finding a significant place in academic settings across various countries. In the context of attributing research output to specific countries, certain publications may be jointly credited to multiple countries. For this study, the country affiliation of the first author serves as the determining factor for allocating a publication to a particular country.

Looking at Table 9, we can see a representation of institutions regarding their publication count acknowledging using ChatGPT. University of Toronto in Canada leads with 16 publications, followed closely by major U.S. institutions like Stanford University, Harvard University, Johns Hopkins University, and Cornell University.

Table 10 shows the distribution of authors based on countries that acknowledge the use of ChatGPT in their scientific work.

The data suggest that ChatGPT is most widely used by authors in the United States, with 395 acknowledging their use of the AI model. China and the United Kingdom follow with 91 and 90 acknowledgments, respectively. India and Germany also show many authors, with 68 and 65 authors, respectively. The countries with the fewest authors acknowledging ChatGPT usage from the list are Saudi Arabia and the United Arab Emirates, with 19 and 18 authors, respectively. This table gives us a glimpse into the global usage of ChatGPT, indicating that it's used across various parts of the world, with a strong presence in the United States.

In Table 11, we see a distribution of researchers based on the number of publications they have contributed that acknowledged the usage of ChatGPT. It's intriguing to see researchers from developed and developing countries, like India, Saudi Arabia, and the United States, actively using this AI tool in their scholarly work. Partha Pratim Ray from Sikkim University, India, has the most publications (8) acknowledging the use of ChatGPT.

When you look at the distribution of researchers based on citations (Table 12), interestingly, the list is topped by an AI researcher labeled “Chat GPT” from OpenAI, United States, indicating that the tool's creators are also active users. Researchers from diverse countries, including the

Table 3. Highly cited publications acknowledging usage of ChatGPT: biomedical and Clinical Sciences.

Title	Relevant portions of Acknowledgements	Explicit Content Generation	OpenAI		Acknowledgement in the References	Source Title	Authors	Times cited
			Assistance and Collaboration	ChatGPT Role				
Artificial Hallucinations in ChatGPT: Implications in Scientific Writing	Some of the text within the manuscript has been generated with the aid of ChatGPT and has been put between quotation marks.	Yes	No	No	Yes	Cureus	Alkaissi, Hussam; McFarlane, Samy I (2023)	89
Applicability of ChatGPT in Assisting to Solve Higher Order Problems in Pathology	We would like to thank ChatGPT for providing free service for research purposes. Without their service, this study could not be conducted. For drafting this manuscript, we used ChatGPT to generate some of the contents and edited according to necessity.	Yes	Yes	Yes	Yes	Cureus	Sinha, Ranwir K. et al., (2023)	21
ChatGPT: Is this version good for healthcare and research?	Acknowledgment Interactions with ChatGPT AI chatbot (on ChatGPT AI platform and on Microsoft Bing AI platform) have been reproduced here.	No	Yes	No		Diabetes & Metabolic Syndrome Clinical Research & Reviews	Vaishya, Raju et al., (2023)	20

Table 4. Highly cited publications acknowledging usage of ChatGPT: information and Computing Sciences.

Title	OpenAI					Source title	Authors	Times cited
	Relevant portions of Acknowledgements	Explicit Content Generation	Assistance and Collaboration	OpenAI and ChatGPT Role	Acknowledgement in the References			
The Carbon Footprint of Machine Learning Training Will Plateau, Then Shrink	Christopher Berner, Ilya Sutskever, OpenAI, and Microsoft shared information about GPT-3.	No	No	Yes	Yes	Computer	Patterson, David et al., (2022)	40
This new conversational AI model can be your friend, philosopher, and guide ... and even your worst enemy	We acknowledge OpenAI for providing free public access to the ChatGPT as a part of the research preview, which allowed us to explore the strengths and weaknesses of this powerful conversational AI model.	No	No	Yes	Yes	Patterns	Chatterjee, Joyjit; Dethlefs, Nina (2023)	36
Large pre-trained language models contain human-like biases of what is right and wrong to do	Further, the authors are thankful to Aleph Alpha for very useful feedback and access to the GPT-3 API.	No	No	Yes	Yes	Nature Machine Intelligence	Schramowski et al. (2022)	30

Table 5. Highly cited publications acknowledging usage of ChatGPT: health Sciences.

Title	Relevant portions of Acknowledgements	Explicit Content Generation	Assistance and Collaboration	OpenAI and ChatGPT Role		Acknowledgement in the References	Source title	Authors	Times cited
				Yes	Yes				
Open artificial intelligence platforms in nursing education: Tools for academic progress or abuse?	Thank you to all the software engineers and staff at OpenAI who developed the language model and AI chatbot known as ChatGPT and made it available open access (https://openai.com). Also, a sincere thanks to a personal WhatsApp group (Dragoncast) which stimulated discussion on the use of OpenAI in scientific writing and prompted the human author to write this editorial using the AI chatbot on its potential application in nursing education.	Yes	Yes	Yes	Yes	Yes	Nurse Education in Practice	O'Connor (2022)	123
Can artificial intelligence help for scientific writing?	For the writing of this article, we have not received funds or support from OpenAI, which is not associated with the entire process that led to the preparation of this article. The text, written with the support of the ChatGPT by OpenAI, has however been modified by the human authors, who assume full responsibility for form and content.	Yes	Yes	Yes	Yes	Yes	Critical Care	Salvagno, Michele et al., (2023)	76
Can ChatGPT draft a research article? An example of population-level vaccine effectiveness analysis	The text of this editorial was written by the four human co-authors, while the use of ChatGPT was restricted to generating the boxes, tables and figures that were explained and discussed in the text.	Yes	Yes	No	Yes	Yes	Journal of Global Health	Macdonald, Calum et al., (2023)	37

Table 6. Highly cited publications acknowledging usage of ChatGPT: engineering.

Title	Relevant portions of Acknowledgements	Explicit Content Generation	Assistance and Collaboration	OpenAI and ChatGPT Role	Acknowledgement in the References	Source title	Authors	Times cited
Role of Chat GPT in Public Health	The author acknowledges that some content in this article was partially generated by ChatGPT (powered by OpenAIs language model, GPT-3.5; http://openai.com) to discover the roles that chatGPT can play in public health. The editing was performed completely by the human author.	Yes	No	Yes	Yes	Annals of Biomedical Engineering	Biswas (2023a)	36
Potential Use of Chat GPT in Global Warming	The author acknowledges that this article was partially generated by ChatGPT (powered by OpenAIs language model, GPT-3; http://openai.com). The editing was performed by the author.	Yes	No	Yes	Yes	Annals of Biomedical Engineering	Biswas (2023b)	17
ChatGPT in Colorectal Surgery: A Promising Tool or a Passing Fad?	The author acknowledges that Supplemental Digital Contents of this article were generated by ChatGPT (powered by OpenAI's language model GPT-3.5; http://openai.com). Editing was done by the authors.	Yes	No	Yes	Yes	Annals of Biomedical Engineering	Li, Wenbo et al., (2023)	6

Table 7. Highly cited publications acknowledging usage of ChatGPT: Psychology.

Title	Relevant portions of Acknowledgements	Explicit Content Generation	Assistance and Collaboration	OpenAI and ChatGPT Role	Acknowledgement in the References	Source title	Authors	Times cited
A Chat(GPT) about the future of scientific publishing	The authors would like to acknowledge ChatGPT for providing the “in the style of Charles Dickens” paragraph and for hours of entertainment in trying to get it to mimic Yoda.	Yes	Yes	No	Yes	Brain Behavior and Immunity	Hill-Yardin, Elisa L et al., (2023)	16
Large Language Models and the Reverse Turing Test	Mitra Hartmann noticed the remarkable parallel between LLMs and the Mirror of Erised. Ilenna Jones tested GPT-3 with counterfactuals in real time during a lecture. Discussions at the Telluride Neuromorphic Cognition Engineering Workshop helped shape my perspective on Large Language Models Finally, I thank GPT-3 for sharing its insights in the closing remarks.	No	Yes	Yes	No	Neural Computation	Sejnowski, Terrence J (2023)	13
The Artificial intelligence large language models and neuropsychiatry practice and research ethic	The author used ChatGPT and Notion AI to polish the language and summarize the editorial. All of its responses are either paraphrased with acknowledgment or, to follow its suggestion, directly quoted with in-text citations.	No	Yes	Yes	No	Asian Journal of Psychiatry	Zhong, Yi et al., (2023)	5

Table 8. Geographical distribution of publications acknowledging the usage of ChatGPT.

Name	Publications	Citations
United States	407	2890
United Kingdom	98	1059
China	92	549
India	75	492
Germany	71	356
Canada	57	203
Australia	56	611
Japan	43	62
Spain	37	236
France	32	236

Table 9. Institutions vs. Number of publications acknowledging usage of ChatGPT.

Name	Country	Publications	Citations
University of Toronto	Canada	16	40
Stanford University	United States	15	105
Harvard University	United States	12	67
Johns Hopkins University	United States	11	192
Cornell University	United States	11	193
New York University	United States	10	204
University of Michigan–Ann Arbor	United States	10	25
University of Pennsylvania	United States	10	50
McGill University	Canada	9	33
King Saud University	Saudi Arabia	9	33

Table 10. Distribution of authors based on countries.

Country	Publications
United States	395
China	91
United Kingdom	90
India	68
Germany	65
Japan	43
Spain	35
Italy	29
Saudi Arabia	19
United Arab Emirates	18

United Kingdom, the United States, Jordan, and India, are represented, demonstrating the global impact of AI-assisted writing. Notably, high citation counts are associated with researchers from prestigious organizations like the University of Manchester and Google, suggesting the high-quality research output made possible using generative AI tools.

Table 11. Researcher productivity based on acknowledging usage of ChatGPT according to publications.

Name	Current Organization	Current Organization Country	Publications	Citations
Partha Pratim Ray	Sikkim University	India	7	30
Himel Mondal	All India Institute of Medical Sciences	India	6	36
Mohamad-Hani A Temsah	King Saud University	Saudi Arabia	6	33
Ayman A Al-Eyadhy	King Saud University	Saudi Arabia	5	32
Amr A Jamal	King Saud University	Saudi Arabia	5	32
Poulami Majumder	West Bengal University of Technology	India	4	2
Fadi Saleh Aljamaan	King Saud University	Saudi Arabia	4	30
Jamil S Samaan	Cedars-Sinai Medical Center	United States	3	1
Yee Hui Yeo	Cedars-Sinai Medical Center	United States	3	1
Ali Asghar Heidari	University of Tehran	Iran	3	0

Table 12. Researcher productivity based on acknowledging usage of ChatGPT according to citations.

Name	Current Organization	Current Organization Country	Citations	Publications
Chat GPT	OpenAI	United States	123	2
Siobhan O'Connor	University of Manchester	United Kingdom	123	1
Alberto Giovanni Gerli	University of Milan	Italy	76	1
Som Subhro Biswas	University of Tennessee Health Science Center	United States	54	3
Debby R E Cotton	Plymouth Marjon University	United Kingdom	48	1
Peter A Cotton	University of Plymouth	United Kingdom	48	1
Quoc-Viet Le	Google	United States	40	1
Maud Texier	Google	United States	40	1
Himel Mondal	All India Institute of Medical Sciences	India	36	6
Nina Dethlefs	University of Hull	United Kingdom	36	1

Journal patterns

Table 13 shows that “Cureus” leads the list with 88 publications, showing a substantial reliance on ChatGPT by its authors. Other top publishers include scientific journals such as “bioRxiv” and “Annals of Biomedical Engineering.” Interestingly, even preprint platforms like “bioRxiv” and “medRxiv,” where researchers share their findings before being peer-reviewed, show significant usage of ChatGPT in their paper writing.

While “JMIR Medical Education” leads in citations, as seen in Table 14, other high-ranking journals emerge, such as “Radiology” and “The Lancet Digital Health,” even though they have fewer publications that utilize ChatGPT. Notably, the top-tier journal “New England Journal of Medicine”

Table 13. Journals and the corresponding number of publications acknowledging usage of ChatGPT.

Journal Name	Publications	Citations
Ranked based on Publications		
Cureus	88	229
Annals of Biomedical Engineering	27	133
bioRxiv	27	11
medRxiv	11	21
JMIR Medical Education	10	299
Sustainability	9	37
IEEE Access	9	8
Scientific Reports	8	18
Aesthetic Surgery Journal	8	72
Library Hi Tech News	7	137

also appears in the list, hinting at the influence of AI-assisted writing even in high-impact scholarly discourse.

Temporal trends

Figure 2 exhibits an overall upward trend in the number of publications acknowledging the use of ChatGPT over time, from January to July 2023. Starting with only 38 publications in January, there is a noticeable leap in March with 152 publications. April and May maintain this heightened output level at 216 and 259 publications, respectively. Then, in June, there was an increase to 291 publications, marking the highest usage of ChatGPT so far. Interestingly, the trend declined slightly in July to 196 publications, possibly due to the timing of indexing of the papers in the database. This temporal evolution indicates a growing acceptance and reliance on ChatGPT as a tool in academic writing. The sharp increase in its usage over a short period suggests that more researchers are becoming aware of its capabilities and integrating it into their work processes. Given this trend, it’s reasonable to expect the usage of ChatGPT in academic writing to continue to rise.

Table 14. Journals and the corresponding number of citations acknowledging usage of ChatGPT.

Journal Name	Citations	Publications
Ranked based on Citations		
JMIR Medical Education	299	10
Cureus	229	88
Radiology	286	6
The Lancet Digital Health	275	3
Healthcare	202	2
JAMA	191	4
New England Journal of Medicine	180	1
International Journal of Information Management	178	1
Journal of Applied Learning & Teaching	171	4
Library Hi Tech News	137	7

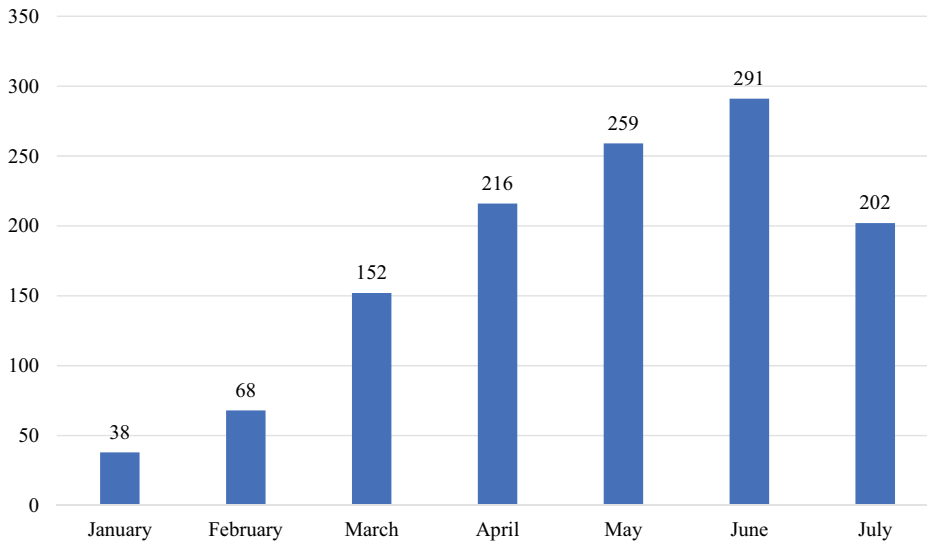


Figure 2. Publications acknowledging the usage of ChatGPT over time.

Conclusions

This study offers an important look into how openly authors admit to using generative AI tools, like ChatGPT, for academic writing across regions of the world. We also understand how these tools are acknowledged worldwide and across different institutions, giving us a snapshot of how AI is being used in scientific research.

From the point of view of different study areas, we see that ChatGPT is being used in various fields, from Biomedical and Clinical Sciences to Engineering and social sciences like Psychology, Education, Communication and Culture, and Business. These fields all show a notable uptake of ChatGPT for creating content, working together, and highlighting the role of AI, showing how varied and widely accepted the tool is. When we look at the world map, the United States is leading the pack in admitting to using ChatGPT in scientific papers, both in the number of publications and the number of authors. But it's not just countries traditionally seen as tech leaders getting on board. There's significant use also happening in countries like China and India, which suggests that generative AI tools are becoming widespread in academic writing, regardless of a country's economic status.

The study also reveals that a diverse mix of educational institutions worldwide are using ChatGPT for their papers. Although institutions in the U.S. are the most frequent users, universities from countries like Saudi Arabia and Canada also make notable use of the tool. Turning to the publishing world, the study shows some interesting patterns. The journal

“Cureus” stands out for having the most ChatGPT publications. However, high-impact journals like “JMIR Medical Education,” “Lancet Digital Health,” “JAMA,” “New England Journal of Medicine” also show significance, hinting at the tool’s impact in scholarly conversations. And it’s not just traditional journals that are benefiting from ChatGPT. Preprint platforms like “bioRxiv” and “medRxiv” also show much usage, pointing to AI’s role in speeding up the sharing of research results without formal peer reviews.

Looking at how usage has changed over time, there’s a clear increase in the use of ChatGPT. This suggests that the tool is being accepted and included more in academic writing practices in a short amount of time. This matches the larger story of AI becoming a bigger part of our everyday lives, and its influence is spreading across different sectors, including academia. One of the interesting things about this study is that it shows that the use of generative AI tools is not confined to established institutions or researchers. Their usage is spreading across countries, disciplines, and publication types, which suggests that AI tools like ChatGPT are starting to become a regular part of the academic world.

The current study does not specifically address whether the researchers using these technologies are ESL (English as a Second Language) or EFL (English as a Foreign Language) writers. However, recognizing the importance of this demographic information, future research could be designed to capture these specifics, potentially offering further insights into the usage patterns and efficacy of these tools among non-native English speakers. ChatGPT and similar generative AI tools do demand considerable computational resources, which might restrict their usability in regions with limited computational infrastructure. Nevertheless, the proliferation of cloud-based services are likely to help with some of these constraints.

As a concluding recommendation, it is pertinent to explore the relationship between artificial intelligence and human users beyond the scope of ChatGPT. In this regard, Pentina et al. (2023) developed a comprehensive literature review that investigated consumer-machine interactions in the contemporary landscape of artificial intelligence. The study revealed an expanding corpus of research dedicated to AI-powered writing aids, including grammar checkers and automated essay scoring systems. Such tools not only assist in enhancing grammatical accuracy, spelling, and writing style but also offer substantive feedback on content. Therefore, the utility of artificial intelligence in writing assistance extends beyond chat-based models like ChatGPT, warranting further scholarly attention.

For the future, it would be valuable to understand how the acknowledgment of AI tool usage might impact those research papers, including aspects like citation rates, peer review outcomes, or public perception in social media platforms. There is a need for studies that investigate the ethical implications of using AI tools in academic research. Such

research can inform the development of robust ethical guidelines for AI tool usage.

However, this study does have some limitations. Firstly, it only covers seven months with a dataset of 1226 papers, so it might not fully represent how ChatGPT would be adopted over a longer period. We might have missed some subjects, places, or smaller institutions using AI tools like ChatGPT. Secondly, it assumes that all authors who used ChatGPT or similar tools have acknowledged their usage, which might not be the case, thus potentially leading to underestimation. Lastly, due to the nature of the data, it is not feasible to validate every acknowledgment claim individually. Hence, the analysis relies on the assumption that the declarations made in the papers are accurate.

So, what does this all mean? This study gives us valuable insights into how AI tools are used in writing scientific papers. The fact that they are being used across different disciplines, in different countries, and more over time suggests that we are moving toward a future where AI plays a larger role in academic research. These trends could lead to more open and collaborative research but also mean we must keep discussing the ethical and practical aspects of using AI in academic settings. This study's findings could benefit a wide array of stakeholders: researchers can track current trends and practices, journal publishers can assess the efficacy of their AI tool usage declaration policies, and institutions can measure their standing in adopting this emerging research tool.

There is a need for clear guidelines and policies on using AI in scientific writing to address ethical considerations. Researchers should be encouraged to disclose AI tools in their work, similar to other methodological procedures or tools. Transparency about AI usage is essential for replicability and honesty and for providing a clearer understanding of the evolving landscape of scientific writing.

Moreover, organizations should provide resources and training on the ethical use of AI in research. Familiarity with the potential of these tools, coupled with an understanding of their limitations, can help researchers effectively and ethically incorporate AI in their work.

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Data availability statement

The dataset used in the study will be made available based on the request.

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